Anticipate or react? Mouse chases measured against the pursuit game

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Abstract

A pursuer can run at where its target is or at where the target is going to be. Pursuit theory names the two answers: pure pursuit is reactive, lead pursuit and parallel navigation are anticipatory. The laws have been fitted to birds, bats and insects but not to rodents, and no study has inferred switching between laws within a chase with its uncertainty. A hidden Markov model over a small library of guidance laws is built here, validated on simulated chases where the truth is known, and fitted to open recordings of mice chasing crickets (485 chases from 7 animals, with a replication set of 177 chases) and of mice chasing mice (13,233 chases). Mouse steering toward a target, in prey capture and in mouse-mouse chases, is intermittent reactive pursuit: a pure-pursuit state with a delay of 33 to 67 ms and a time constant of 0.15 to 0.18 s holds 15 to 23 percent of a chase's frames on average by decoded state, the anticipatory laws collapse onto it, and the remaining frames show no response to the target. A generic autoregressive switching model with no reference to the target explains the primary dataset's head yaw as well as the law library does, and explains the conspecific body heading far better, so the evidence for pursuit laws beyond intermittent reactive steering is thin. Measured against the pursuit game, target manoeuvres do not depend on the pursuer's state, and the chased mouse does not steer away from the chaser in any form the evader library expresses. These statements apply to frames in which the target moves, a minority of frames in the prey-capture data.

Keywords: differential games; pursuit-evasion games; hidden Markov models; model comparison; guidance laws; proportional navigation; sensorimotor control; prey capture; mouse behaviour.
Mathematics Subject Classification: 91A23, 49N75, 62M05, 92D50.

1 Introduction

A mouse closing on a cricket faces a choice it never states. The cricket sits still, the mouse runs at it, and then the cricket jumps. The mouse can turn toward where the cricket now is, or toward where the cricket will be a moment from now if it keeps going. The first choice closes a loop on current information. The second commits to a prediction. Every chaser faces the same choice at every moment, and the two answers leave different marks on the chaser's path: a reactive chaser keeps its heading on the target and follows a curved path behind it, an anticipatory chaser aims ahead of the target and cuts the corner.

Pursuit theory has named these answers for a long time. Pure pursuit points the chaser at the target's current position. Lead pursuit points it at the target's position extrapolated by a lead time. Parallel navigation, the classical interception law, turns the chaser so that the bearing to the target stays constant, which is also the pursuer's optimal strategy in the simplest pursuit game [1–4]. These laws have been fitted to the attacks of peregrine falcons and hawks [5, 6], to

echolocating bats [7, 8], to dragonflies [9, 10] and to people intercepting a target on foot [11, 12]. Mice hunt crickets under visual guidance [13–17] and pursue one another; pursuit has been read in terms of the target's position, velocity and acceleration in macaques [18] and in rats [19, 20], and recurrent networks trained on the task come to anticipate the target [21]. No study has fitted the guidance laws to a rodent, and no study of any animal has inferred switching between laws within a chase while quantifying its uncertainty.

What is missing is a way to tell, moment by moment, which answer the animal is giving, with honest uncertainty. The method built here is a hidden Markov model over a small, nested library of guidance laws with a shared motor delay, gain and noise model, fitted by expectation-maximisation with a forward-backward pass over every chase, and validated on simulated chases where the generating law and its switches are known. It applies the idea of a hidden state that selects a sensorimotor rule [22, 23] to a library of guidance laws rather than to pose syllables [24, 25] or to laws of motion without a target [26]. Applied to open recordings of mice chasing crickets and mice chasing mice, and measured against the pursuit game through a distance from optimal play and a test of whether the target responds to the pursuer, it gives a plain answer. Mouse steering toward a target is intermittent reactive pursuit: a pure-pursuit state with a delay of one to two frames and a time constant of 0.15 to 0.18 s holds 15 to 23 percent of a chase's frames on average by decoded state, the anticipatory laws collapse onto it, and the remaining frames show no response to the target.

2 Results

2.1 The library, the model and what it recovers on simulated chases

Three guidance laws map the current geometry (pursuer position and heading, target position and velocity) to a commanded heading change (Figure 1). Pure pursuit turns the pursuer toward the target's current position. Lead pursuit turns it toward the target's position extrapolated by a lead time τ, and contains pure pursuit at $\tau = 0$. Proportional navigation turns the pursuer in proportion to the rotation of the line of sight with a navigation constant N; parallel navigation, the interception law, is its large-N limit. The laws share a motor delay d in frames, a steering gain and a noise model on the heading increment, so that the comparison between them concerns the direction the pursuer is sent in and nothing else. Lead pursuit with $\tau > 0$ and proportional navigation are anticipatory in the sense used throughout: both send the pursuer toward where the target will be if it keeps its current velocity. The model sees that velocity and nothing richer, so anticipation here never means prediction of a manoeuvre.

A hidden Markov model over the law index, with plain or input-dependent transitions, gives at every frame the posterior probability of each law. On a suite of simulated chases whose conditions span noise scales from 0.02 to 0.20 rad per frame and chase lengths of 30, 60, 120, 180 frames, with the generating law and its switches known, the model classifies held-out single-law chases correctly in 0.95 to 1.00 of cases at full length (Figure 2A), detects a scripted switch in 0.95 to 1.00 of chases (a Markov switch in 1.00 to 1.00, a range-triggered switch in 0.95 to 1.00) against a null set on single-law chases with and without tracking noise, and raises no false switch on misspecified targets (evaders and jumpers) above a rate of 0.02. The decoded switch time has a median error of 2 frames, with 86 percent of decoded switches within five frames. Bootstrap intervals cover the lead time in 1.00 of conditions, the navigation constant in 0.92 and the gain in 0.83; the delay is recovered exactly in 1.00 of 15 conditions over a grid of zero to eight frames. The posterior is calibrated to a weighted gap of 0.013 between stated probability and empirical frequency (Figure 2D).

Two limits bound what follows. A reactive switch after a target jump is not identifiable when the target stops after the jump, because every law then commands the same heading (detection 0.00 against 1.00 when the target keeps moving). And at the gains, delays and noise of the real recordings (a realistic block of the suite with a generating gain of 0.05 per frame, recovered as

0.051), per-chase classification of the law from a single chase is unreliable: 0.70 of 60 held-out chases are classified correctly at full length and 0.49 to 0.50 at 30 to 60 frames, with a still target 0.45, and per-chase switching detection against the realistic null is at chance (0.10). The real-data claims below therefore rest on pooled held-out likelihoods and on a parametric null at the dataset level, not on per-chase law assignments.

2.2 Datasets

The primary dataset is the overhead tracking of 7 mice hunting live crickets [27]: head position, head yaw and cricket position at 30 frames per second. Of 105 clips, 7 were excluded because their yaw track flips by more than one radian between consecutive frames on more than one percent of frame pairs, a rate of head rotation a mouse does not produce (Methods). The shared segmentation rule gives 485 chases, split by animal into a development set (210 chases, 4 animals) on which every design choice was made and a held-out set (275 chases, 3 animals) scored once. The cricket moves faster than 3 cm/s on 11 percent of held-out frames (1,022 of 9,571); these informative frames are the ones that separate anticipatory from reactive steering, and every comparison is reported on all frames and on them. The replication set is the tracking of 3 mice hunting crickets with olfactory and auditory cues masked [28], 177 chases at 28.6 frames per second after decimation, fitted once with the frozen model; its cricket moves on 38 percent of frames. The conspecific dataset is the MABe22 mouse-triplet tracking [29]: 13,233 chases of one mouse by another by the same rule, split by recording file into 5,485 development and 7,748 held-out chases, with the target moving on 65 percent of frames. Table 1 lists the splits.

2.3 Law comparison on held-out chases

Figure 3 and Table 2 give the held-out log-likelihood per frame of each model minus that of the best single law, with chase-level bootstrap intervals. In the primary dataset the best single law is pure pursuit with a gain of 0.021 per frame; the persistence null sits 0.0086 nats per frame below it (interval -0.0118 to -0.0054): the laws explain something, but little. At the single-law gain the lead time is not identified, because the lead term it multiplies is too small to move the likelihood, so lead pursuit ties with pure pursuit (difference 0.0000); proportional navigation alone sits 0.0075 below pure pursuit. The replication set gives the same ordering (best single lead pursuit, tied with pure pursuit at 0.0000; the null 0.0125 below), and so does the conspecific set (the null 0.0017 below, lead pursuit -0.0006, proportional navigation -0.0016). On informative frames the ordering is unchanged in every dataset (Table 2, right columns).

2.4 Switching: two states, one of them not steering

The switching model beats the best single law on held-out chases in every dataset: by +0.0231 nats per frame (+0.0173 to +0.0294) in the primary set, +0.0255 (+0.0184 to +0.0327) in the replication set and +0.0768 (+0.0749 to +0.0786) in the conspecific set. Against the parametric null, in which the best single law is replayed along the recorded target paths with tracking noise and the whole comparison repeated, the observed gaps exceed every one of 200, 200 and 200 replicates (null 95th percentiles -0.0007, -0.0003 and -0.0011; $p < 0.005$, < 0.005 and < 0.005, the smallest value 200 replicates allow). Input-dependent transitions add +0.0309, +0.0351 and +0.0788 over the best single law; the gain over fixed transitions is set out below.

The model was given three laws, but it uses them as two behaviours (Table 3). The pure-pursuit state carries a gain of 0.188 per frame (0.152 to 0.292; 0.238 in the replication set, 0.221 in the mice), a time constant of 0.18 s at a delay of 1 frame (33 ms; 35 ms in the replication set, 67 ms in the mice). The lead-pursuit state collapses onto it, with a lead time of 0.001 s (0.000 to 0.344). The proportional-navigation state fits a navigation constant of -0.017 (-0.067 to 0.112;

0.046 and -0.003 in the other sets), which commands no heading change: it is a state in which the pursuer does not steer toward the target. Within the decoded frames of that state the heading increment is unrelated to the bearing, while within the pure-pursuit frames it follows the bearing (Methods). States are therefore named by what they do: a steering state (pure pursuit, and any law with a non-zero response) and a non-steering state, and the reported posterior is the steering probability. Across held-out frames the mean steering probability is 0.40 in the primary set (0.39 on informative frames), 0.25 in the replication set and 0.29 in the mice; by decoded path the steering state holds 23, 15 and 23 percent of a chase's frames on average. Figure 4 shows three held-out chases of the primary set with the posterior underneath.

What triggers a switch? Transitions that depend on range, closing speed and a recent target manoeuvre improve the held-out fit over fixed transitions by +0.0078 (+0.0034 to +0.0119), +0.0096 (+0.0067 to +0.0128) and +0.0021 (+0.0019 to +0.0023) nats per frame: 34 and 38 percent of the switching gain in the two cricket sets and 3 percent in the mice (Table 13). Which covariate carries the improvement is not resolved here. The steering probability aligned to the onset of target manoeuvres (Figure 5) is flat in the replication set (0.29 in the half second before a manoeuvre, 0.27 in the half second after) and too noisy to read in the primary set, where 17 chases contain a manoeuvre. In the mice it falls from 0.36 to 0.24. Steering declines over the course of a mouse-and-mouse chase and manoeuvres come late, and a prediction from chase phase alone falls from 0.32 to 0.24 over the same lags, 64 percent of the observed fall. The lead time refitted within tertiles of target speed and range stays at or below 0.02 s in the primary set and 0.12 s in the replication set, and between 0.21 and 0.29 s in the mice, where the lead state holds 1 percent of frames (Figure 6C).

2.5 A generic comparator

A switching model with no law library, three autoregressive states on the heading increment with the same noise model as the law models, tests whether the library adds anything beyond a generic segmentation of the steering signal. In the primary set it matches the law model: +0.0283 nats per frame over the best single law (+0.0197 to +0.0373) against +0.0231 for the law model, with overlapping intervals. In the replication set the law model wins (+0.0059 against +0.0255). In the conspecific set the comparator wins by a wide margin, +0.2957 against +0.0768: the body-axis heading of a mouse has turn dynamics of its own, smooth enough that the shared residual autocorrelation sits at its bound (0.95), and a model of those dynamics explains the heading better than any law plus a first-order residual. The law library is not the best description of what a chasing mouse's heading does; it is a description of the part of it that points at the target.

2.6 The follow-up with a persistence state

Because the frozen library had no state for not steering and borrowed one, a pre-declared follow-up after the freeze added an explicit persistence state (zero predicted increment) as a fourth state, with the comparator given four states as well, and asked whether proportional navigation then takes a non-zero constant (Table 6). With the persistence state carrying the non-steering frames, the navigation constant is -0.025 (-0.073 to 0.117) in the primary set, 0.158 in the replication set and -0.547 in the mice; the persistence state holds 33, 53 and 69 percent of decoded frames, and the four-state model's gap over the best single law is +0.0276, +0.0292 and +0.0829 nats per frame, against +0.0231, +0.0255 and +0.0768 for the frozen three-state model. Freed of its non-steering duty, proportional navigation takes a constant distinguishable from zero only in the replication set, and a small one: a constant of 0.158 turns the pursuer by that fraction of the line-of-sight rotation, far from the constant of three or more that intercepts a target. In the mice the state shrinks to 3 percent of decoded frames with a negative constant, a turn against the line-of-sight rotation, which is not a guidance law and reads as the model absorbing a few

frames of large heading change. The follow-up is reported after the frozen analysis and does not replace it; it does not change the answer.

2.7 Distance from optimal play

In the simple-motion pursuit game, a differential game in Isaacs' sense whose saddle-point strategies are known in closed form, the pursuer's optimal strategy holds the bearing constant and the evader runs straight away along the line of sight [1, 2]. The distance from optimal play of a chase is the time-averaged absolute angle between the pursuer's adopted heading and the constant-bearing course given the same geometry (Figure 7, Table 5). Its median is 23° in the primary set (32° on informative frames), 31° in the replication set and 64° in the mice, against 20°, 25° and 44° from the pure-pursuit heading: the mouse's heading sits closer to pure pursuit than to the collision course in every dataset. How the distance varies differs between datasets. In the primary set it grows with range (from 22° in the closest fifth of frames to 42° in the farthest) and with the phase of the chase (30° to 38°); in the replication set it is larger at long range (30° to 42°) with no consistent trend over the chase; in the mice it grows with phase (40° to 100°) but not with range (75° to 64°). Against the recorded target path, the optimal pursuer at the mouse's own speeds would have captured in a median of 1.01 s where the mouse took 1.26 s in the replication set (55 chases ending in contact; ratio 1.15), and 1.03 against 1.30 s in the mice (451 chases; ratio 1.12). The primary held-out set has only 5 chases ending in contact.

2.8 The two-player test

Treating the target as a player, a logistic regression of target-manoeuvre onsets on range, closing speed and the pursuer's steering probability at the previous frame, with chase-level cluster bootstrap intervals, asks whether manoeuvres depend on the pursuer's state (Figure 8A, Table 5). The null circularly shifts the steering covariate within each chase and so keeps the between-chase association; the test statistic is the distance of the observed coefficient from the centre of the null distribution. The within-chase effect is -0.55 (permutation $p = 0.275$, 33 onsets) in the primary set, -0.15 ($p = 0.555$, 268 onsets) in the replication set and +0.02 ($p = 0.693$, 5,491 onsets) in the mice; manoeuvres are more likely when the pursuer is close in every dataset (coefficients of -2.72, -0.38 and -0.10 per 10 cm of range). For the mouse targets, the evader library (pure evasion, lead evasion, straight flight) fitted to the chased mouse's own steering gives an evasion gain of -0.0006: the chased mouse does not steer away from the chaser in any form the library expresses, its three evader states are not distinguishable, the cross-correlation between the pursuer's steering probability and the evader's lead-evasion probability is +0.011 at zero lag and inside the chase-shuffle null at every lag (Figure 8B), and the joint occupancy of pursuer state by evader mode (Figure 8C) carries no information beyond the pursuer's own state.

2.9 Predator-prey against conspecific chases, and two sensitivity checks

Figure 9 places the three datasets side by side: the steering probability per chase, the held-out evidence for switching, and the observed switching gap against its null. The structure is the same in the cricket and the mouse chases, intermittent reactive pursuit with a non-steering state, and the differences are of degree: a longer delay in the mice (67 against 33 ms), a residual far smoother in the body heading than in the head yaw, and a comparator that overtakes the law library on the body heading. Two checks bear on the primary results. Scored leave-one-animal-out at the frozen delay, the replication set gives a switching gap of +0.0218 nats per frame (+0.0141 to +0.0296) and a comparator gap of +0.0050, against +0.0255 and +0.0059 in sample (Table 11). At an arena-fit scale of 24.6 by 20.4 px/cm instead of the deposit's 27, which changes which chases and frames pass the thresholds, the primary dataset gives a gain of 0.168 per frame and a switching gap of +0.0207 (+0.0160 to +0.0264; Table 12).

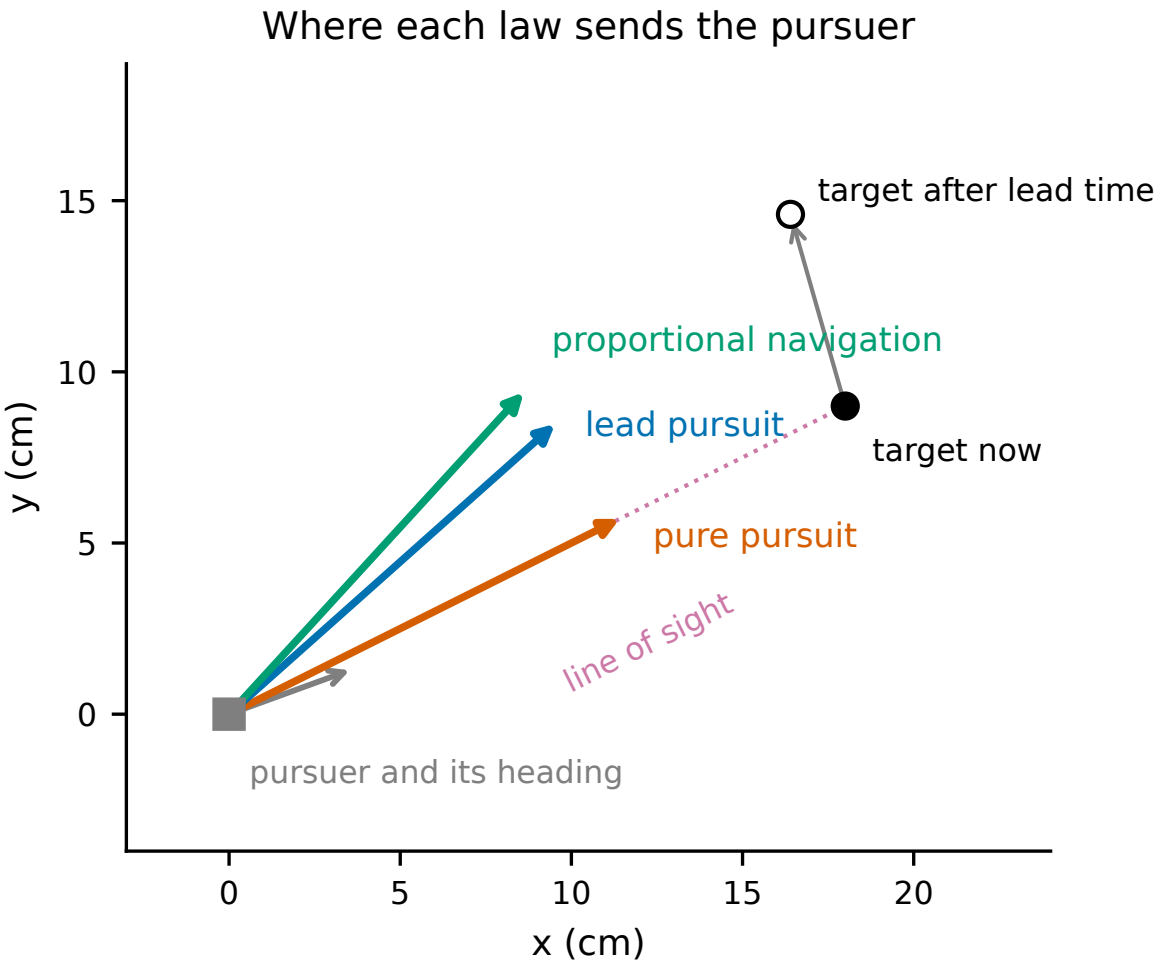


Figure 1: The three guidance laws on one geometry. Where each law sends the pursuer (grey square) given the target (black dot) and its velocity: pure pursuit at the target, lead pursuit at the target extrapolated by the lead time, proportional navigation along the course that holds the bearing.

Table 1: Datasets and splits. Informative frames are those with the target faster than 3 cm/s. For the mouse-and-mouse data, mouse identities are not linked across one-minute sequences, so the animal count is the count of mice per sequence.

dataset	split	chases	animals	frames	moving frames	median s	fps
mouse and cricket (primary)	dev	210	4	7,713	748	0.97	30.00
mouse and cricket (primary)	held	275	3	9,571	1,022	0.93	30.00
mouse and cricket (replication)	replication	177	3	8,245	3,125	1.29	28.60
mouse and mouse	dev	5,485	1,987	286,947	193,415	1.47	30.00
mouse and mouse	held	7,748	2,767	400,499	261,207	1.47	30.00

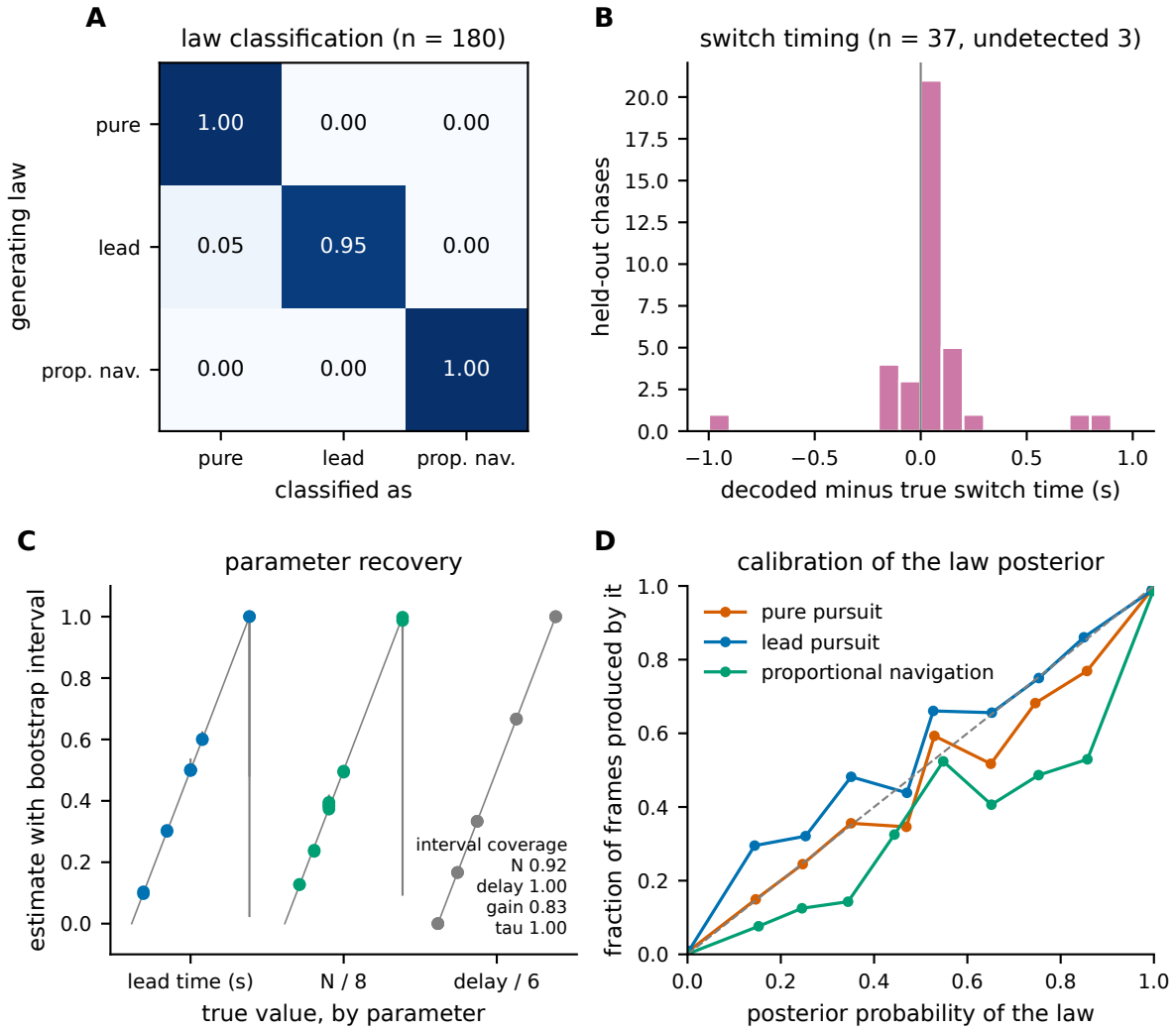


Figure 2: Recovery on simulated chases. (A) Classification of held-out full-length single-law chases by the single-law likelihoods, pooled over noise levels up to 0.1 rad per frame. (B) Decoded switch time against the scripted one. (C) Recovered lead time and navigation constant with bootstrap intervals against the generating values. (D) Calibration of the posterior law probabilities.

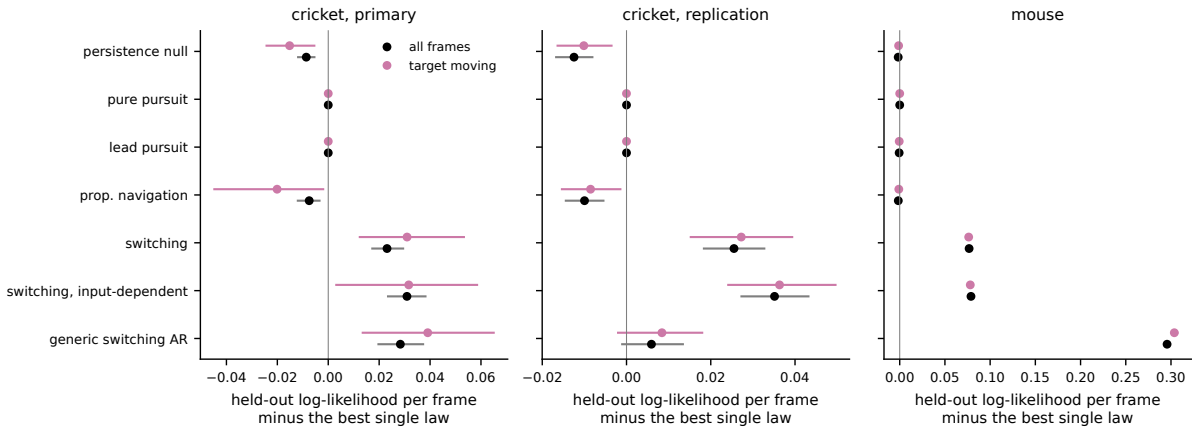


Figure 3: Model comparison on held-out chases. Held-out log-likelihood per frame of each model minus that of the best single law, with chase-level bootstrap intervals, on all frames (black) and on frames with the target moving faster than 3 cm/s (pink), per dataset. All models share the delay, the autoregressive residual and the outlier component; the generic switching autoregressive comparator shares them and has no law library.

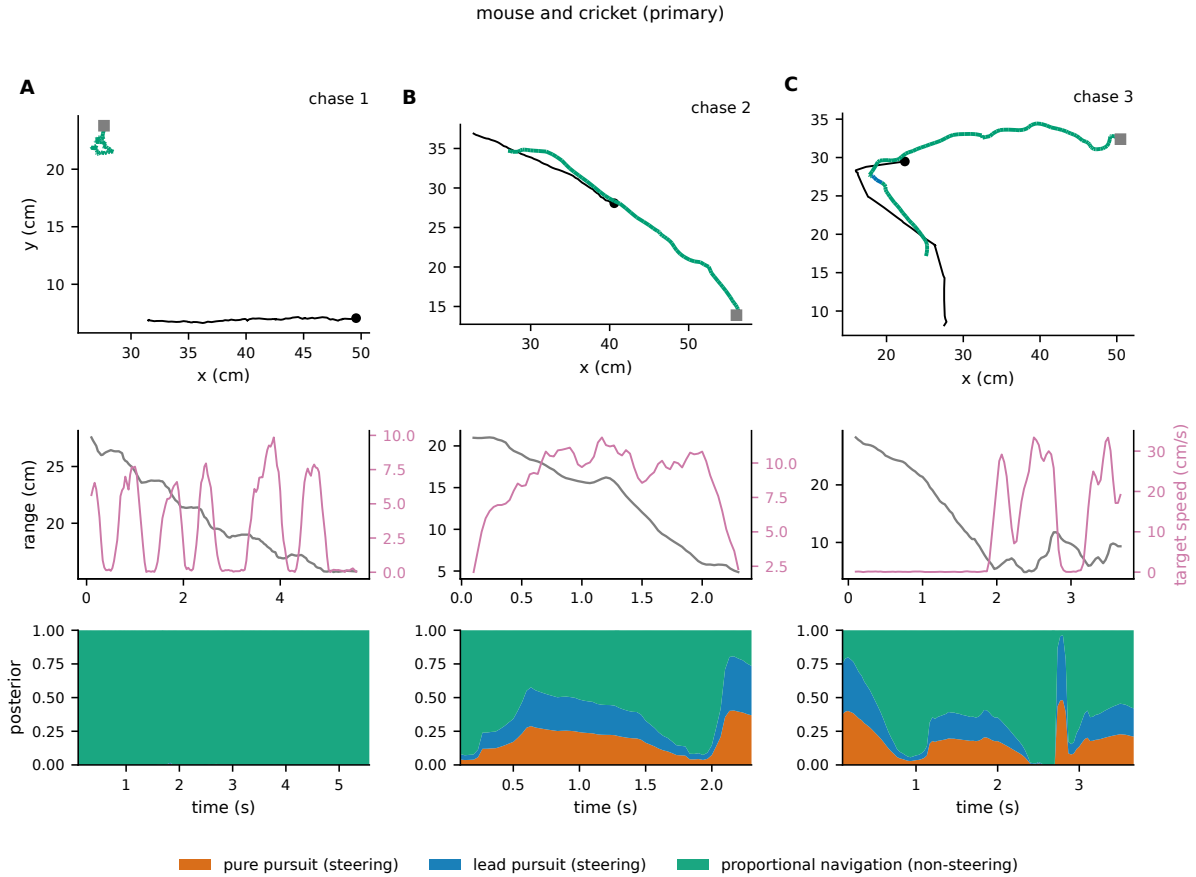


Figure 4: Three held-out chases of the primary dataset. Top: the cricket's path (black) and the mouse head's path coloured by the decoded state: orange pure pursuit and blue lead pursuit are the steering states; green is the state given proportional navigation, whose navigation constant is zero, the non-steering state. Middle: range and target speed. Bottom: posterior probability of each state.

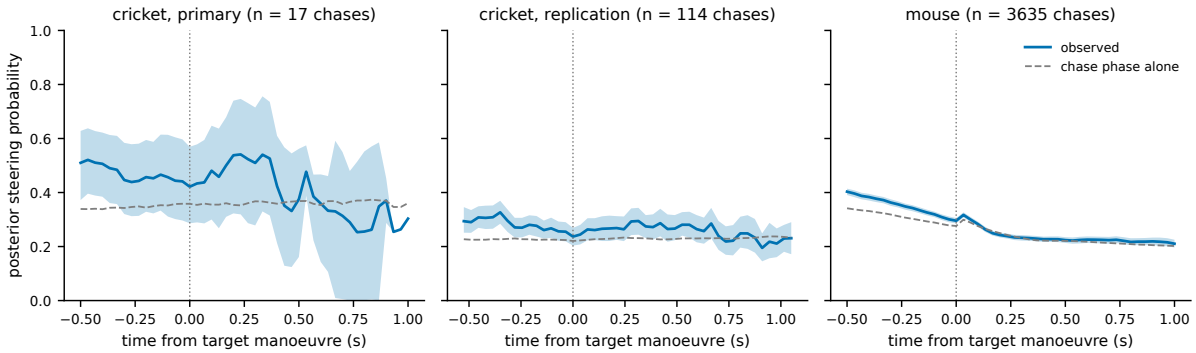


Figure 5: Steering probability around target manoeuvres. Mean posterior steering probability aligned to the onset of a target manoeuvre, with a chase-level bootstrap band, per dataset; dashed, the prediction from chase phase alone. It is flat in the replication set and noisy in the primary set (17 chases); in the mice it falls across the manoeuvre, mostly with the decline of steering over the course of a chase (Results).

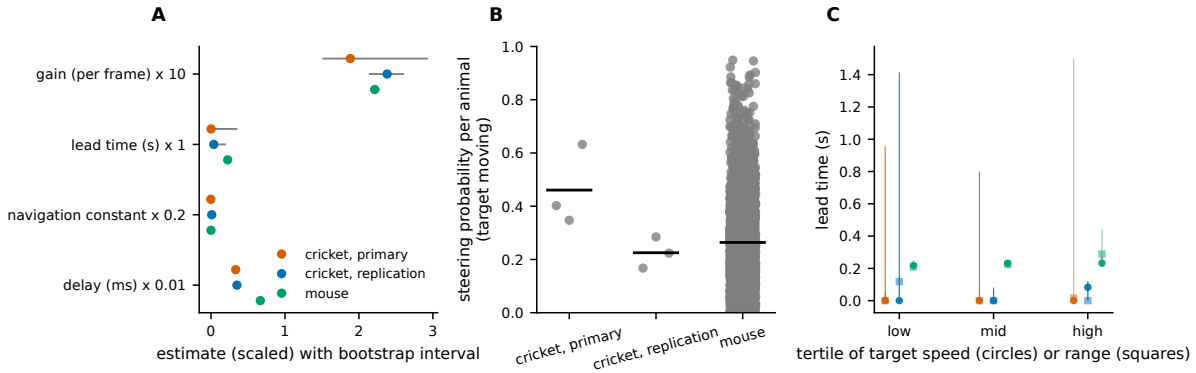


Figure 6: Fitted parameters. (A) Gain, lead time, navigation constant and delay with chase-level bootstrap intervals, per dataset. (B) Steering probability per animal on frames with a moving target. (C) Lead time refitted within tertiles of target speed (circles) and range (squares), coloured by dataset as in A.

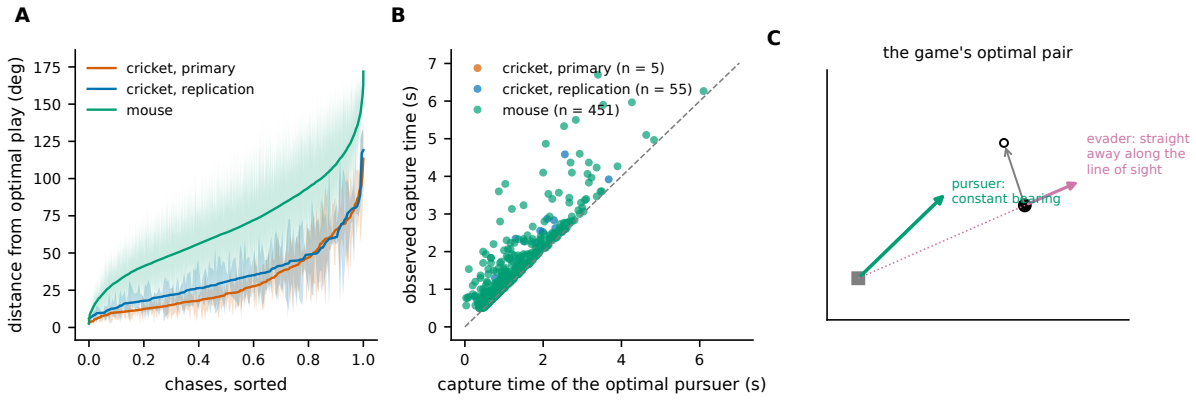


Figure 7: Distance from optimal play. (A) Per-chase distance from the constant-bearing course with block-bootstrap intervals, chases sorted. (B) Observed capture time against the capture time of the optimal pursuer along the same target path at the observed speeds, for chases ending in contact. (C) The optimal pair of strategies of the simple-motion game.

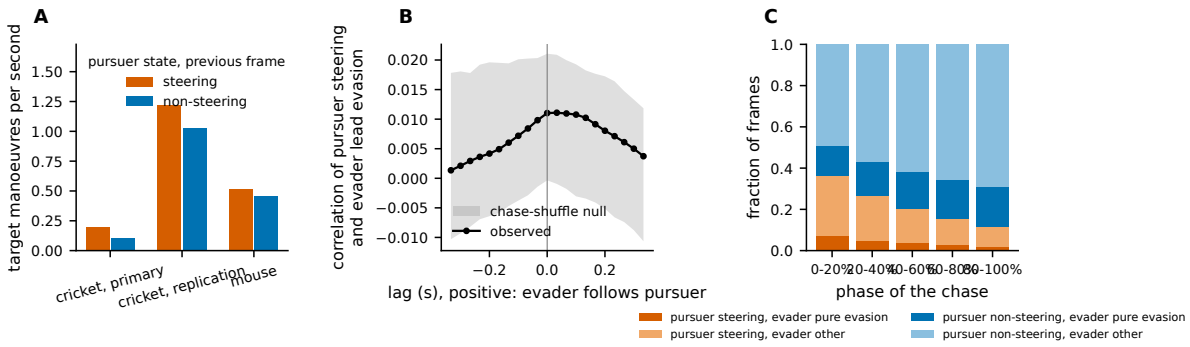


Figure 8: The two-player test. (A) Target manoeuvre rate when the pursuer's decoded state at the previous frame was steering or non-steering; raw rates, before the adjustment for range and closing speed and the within-chase comparison of the test. (B) Mouse-and-mouse chases: cross-correlation between the pursuer's steering probability and the evader's lead-evasion probability, with the chase-shuffle null band. (C) Joint occupancy of pursuer state by evader mode across fifths of the chase.

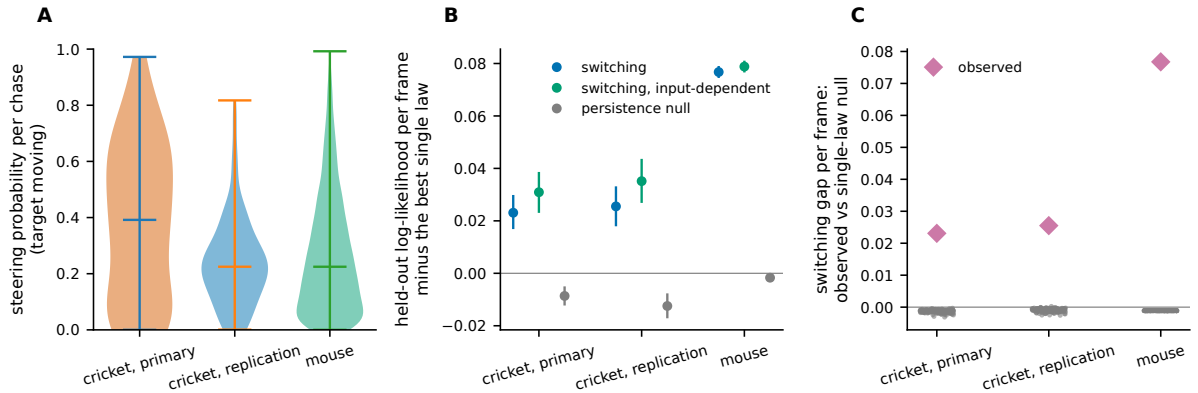


Figure 9: Predator-prey against conspecific chases. (A) Steering probability per chase (target moving). (B) Held-out evidence for switching relative to the best single law. (C) The observed switching gap against its parametric-bootstrap null.

Table 2: Held-out log-likelihood per frame of each model minus the best single law, with chase-level bootstrap intervals, on all frames and on informative frames.

dataset	model	per frame	diff	lo	hi	diff, moving	lo	hi
mouse and cricket (primary)	null	1.4237	-0.0086	-0.0118	-0.0054	-0.0152	-0.0242	-0.0055
mouse and cricket (primary)	single pure pursuit	1.4324	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
mouse and cricket (primary)	single lead pursuit	1.4324	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
mouse and cricket (primary)	single proportional navigation	1.4249	-0.0075	-0.0119	-0.0034	-0.0201	-0.0447	-0.0020
mouse and cricket (primary)	switching plain	1.4554	0.0231	0.0173	0.0294	0.0310	0.0124	0.0533
mouse and cricket (primary)	switching input	1.4633	0.0309	0.0235	0.0382	0.0316	0.0032	0.0585
mouse and cricket (primary)	switching ar	1.4607	0.0283	0.0197	0.0373	0.0391	0.0136	0.0650
mouse and cricket (replication)	null	0.9042	-0.0125	-0.0167	-0.0081	-0.0101	-0.0164	-0.0036
mouse and cricket (replication)	single pure pursuit	0.9166	0.0000	0.0000	0.0000	0.0000	-0.0001	0.0001
mouse and cricket (replication)	single lead pursuit	0.9166	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
mouse and cricket (replication)	single proportional navigation	0.9067	-0.0100	-0.0144	-0.0055	-0.0085	-0.0153	-0.0015
mouse and cricket (replication)	switching plain	0.9421	0.0255	0.0184	0.0327	0.0272	0.0153	0.0393
mouse and cricket (replication)	switching input	0.9518	0.0351	0.0273	0.0432	0.0363	0.0242	0.0496
mouse and cricket (replication)	switching ar	0.9226	0.0059	-0.0010	0.0134	0.0084	-0.0020	0.0180
mouse and mouse	null	2.6269	-0.0017	-0.0019	-0.0014	-0.0012	-0.0016	-0.0008
mouse and mouse	single pure pursuit	2.6286	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
mouse and mouse	single lead pursuit	2.6280	-0.0006	-0.0008	-0.0004	-0.0005	-0.0009	-0.0002
mouse and mouse	single proportional navigation	2.6270	-0.0016	-0.0018	-0.0013	-0.0010	-0.0014	-0.0007
mouse and mouse	switching plain	2.7053	0.0768	0.0749	0.0786	0.0763	0.0739	0.0786
mouse and mouse	switching input	2.7074	0.0788	0.0770	0.0806	0.0780	0.0759	0.0804
mouse and mouse	switching ar	2.9243	0.2957	0.2933	0.2980	0.3037	0.3007	0.3070

Table 3: Fitted parameters of the switching model on the development sets with chase-level bootstrap intervals, and the switching gap against its parametric-bootstrap null. Key columns; all columns are in Table 7 in the supplement.

dataset	delay (ms)	gain per frame	gain interval	lead time (s)	navigation constant	switching gap	null p
cricket, primary	33.3	0.188	0.152 to 0.292	0.001	-0.017	0.0231	<0.005
cricket, replication	35.0	0.238	0.215 to 0.260	0.037	0.046	0.0255	<0.005
mouse	66.7	0.221	0.217 to 0.227	0.225	-0.003	0.0768	<0.005

Table 4: Posterior summaries on the held-out chases: steering probability, decoded state shares and switches. Key columns; all columns are in Table 8 in the supplement.

dataset	steering probability	steering, target moving	steering share of frames	switches per chase	chases with a switch
cricket, primary	0.40	0.39	0.23	0.13	0.11
cricket, replication	0.25	0.24	0.15	1.32	0.51
mouse	0.29	0.27	0.23	0.83	0.50

Table 5: Distance from optimal play and the two-player test. Key columns; all columns are in Table 9 in the supplement.

dataset	from optimal (deg)	from pure pursuit (deg)	capture time ratio	within-chase effect	permutation p	manoeuvre onsets
cricket, primary	22.7	20.2	1.21	-0.55	0.275	33
cricket, replication	30.7	25.3	1.15	-0.15	0.555	268
mouse	63.6	44.0	1.12	0.02	0.693	5,491

Table 6: The post-freeze follow-up with a persistence state, against the frozen three-law analysis. Key columns; all columns are in Table 10 in the supplement.

dataset, library	navigation constant	navigation constant interval	persistence share	switching gap	null p
cricket, primary, 3 laws	-0.017	-0.067 to 0.112	0.00	0.0231	<0.005
cricket, primary, + persistence	-0.025	-0.073 to 0.117	0.33	0.0276	0.032
cricket, replication, 3 laws	0.046	-0.014 to 0.114	0.00	0.0255	<0.005
cricket, replication, + persistence	0.158	0.029 to 0.262	0.53	0.0292	0.032
mouse, 3 laws	-0.003	-0.004 to -0.001	0.00	0.0768	<0.005
mouse, + persistence	-0.547	-1.143 to -0.539	0.69	0.0829	0.048

3 Discussion

The question in the title has a plain answer in these recordings: a chasing mouse reacts, and only part of the time. The guidance-law library was built so that anticipation would show up as a lead time above zero or a navigation constant above zero. Neither appears. Lead pursuit collapses onto pure pursuit in every dataset, and the state that was given proportional navigation fits a navigation constant at zero, which is not a law but the absence of one. What the switching model finds instead is intermittency: a steering state in which the head (or the body) turns toward the target with a short delay and a time constant of 0.15 to 0.18 s, holding 15 to 23 percent of a chase's frames on average by decoded state, and a non-steering state in which the heading increment is unrelated to where the target is. The switching between the two is real by every test applied, and transitions that depend on range, closing speed and recent target manoeuvres predict held-out chases better than fixed transitions.

Three cautions frame this. First, the information structure of the model. The laws see the target's current velocity and nothing richer, so anticipation here means acting on a linear extrapolation of that velocity. A mouse that predicted a cricket's jump from its posture, or that aimed at a learnt position, would not be anticipatory in this sense and would be scored as reactive or as not steering. The negative result on anticipation is a result about linear extrapolation of the target's motion. Second, the recovery study bounds the claims. At the gains and noise of the real recordings, a single chase does not identify its law, and a reactive switch after a jump cannot be seen when the target stops; the claims rest on pooled held-out likelihoods, on a parametric null at the dataset level and on the within-state relation between heading increment and bearing, not on per-chase assignments. Third, both prey-capture datasets record unusual hunts: the primary mice wore head-mounted eye cameras and an inertial sensor, and the replication mice hunted with olfactory and auditory cues masked. That the same two-state structure appears in both, and in the unencumbered mice of the conspecific dataset, is some reassurance, but a mouse hunting without hardware and with all its senses might steer differently.

The generic comparator is the result that most constrains interpretation. A switching model with three autoregressive states on the heading increment and no reference to the target explains the head yaw of the primary dataset as well as the law model, and explains the body heading of the conspecific dataset far better. The law model's advantage over the null is the part of the heading that points at the target; the comparator's advantage is the rest, the turn dynamics of a head or a body that moves for its own reasons. A body-axis heading in particular has smooth turn episodes that a first-order residual cannot carry, and the shared autocorrelation sitting at its bound in the mouse data says so. A model that combined the law library with a richer heading dynamics would likely beat both. The follow-up with a persistence state, which asks the narrower question of whether proportional navigation takes a non-zero constant once it no longer has to serve as the non-steering state, is reported with its answer in Table 6.

Measured against the pursuit game, the interaction tested as non-strategic. The mouse's heading sits closer to pure pursuit than to the collision course the game prescribes, and departs from it more at long range in the cricket data and later in the chase in the primary set and the mice. The target's manoeuvres do not depend on the pursuer's state once the between-chase association is removed, and the chased mouse does not steer away from the chaser in any form the evader library expresses. The chase can be read as a pursuit game, and that is the reading that gave the distance from optimal play and the two-player test their definitions, but nothing here solves a game from the data: no payoffs were estimated and no equilibrium is claimed. The information-structure caveat applies to the target as much as to the pursuer.

What would settle the open points is closed-loop control of the target, so that a manoeuvre can be placed at a chosen moment of the pursuer's state and the response measured, which a robot target in an arena of adjustable complexity now allows [30, 31], and neural recordings aligned to the inferred switches between steering and not steering. The method is ready for both: it runs on any tracked chase in the common format, returns a posterior over states at

every frame with a delay and a gain, and reports when it cannot tell.

4 Methods

4.1 Data

Primary dataset. Michaiel, Abe and Niell [15, 27] recorded 7 mice hunting live crickets in a 45 by 38 cm arena with an overhead camera at 30 frames per second while the mice wore head-mounted eye cameras and an inertial sensor; the deposit gives the head centre, the head yaw from DeepLabCut [32] head points, and the cricket centre in one pixel frame, resampled by the authors onto a 60 Hz grid, of which every second sample is used. The scale is the authors' 27 px/cm. The pursuer heading is the head yaw, its sign convention calibrated against the direction of travel at high speed. Single-frame flips of the yaw (a sample more than 90° from both neighbours while the neighbours agree) are replaced by the circular mean of the neighbours; every other sample is used as measured. Clips in which more than 1 percent of consecutive yaw samples differ by more than 1 radian, a rate of head rotation a mouse does not produce, are excluded as untracked: 7 of 105 clips, 24,324 frames, with flip rates of 2 to 8 percent against at most 0.7 percent in the others. The development split is animals J470c, J462c, J475c and J463b; the held-out split is J462b, J462a and J463c.

Replication dataset. Holmgren and colleagues [16, 28] tracked 3 mice hunting crickets with olfactory and auditory cues masked, with head position and orientation and the cricket position at 200 Hz in a centimetre frame; the series are decimated by seven to 28.6 frames per second so that the delay grid in frames spans the same seconds as at 30 Hz. The pursuer heading is the horizontal projection of the head's forward axis. The set is fitted once, in sample, with the frozen structure, and scored leave-one-animal-out as a sensitivity analysis.

Conspecific dataset. MABe22 [29, 33] tracks three mice per recording at 30 frames per second with twelve keypoints per mouse. The pursuer heading is the body axis from tail base to neck, the target position the body centre of the chased mouse. The pixel scale is estimated per file from the extent of the centre-of-back keypoint against the arena size. Mouse identities are not linked across sequences, so each mouse of each sequence is its own stratum in the chase-level bootstrap. The training and submission files form the development split and the test file the held-out split.

Chases. One rule cuts chases from every recording. A chase starts when the pursuer closes on the target faster than 5 cm/s while moving faster than 10 cm/s with the target within 60° of its heading, from a range of at least 8 cm; it ends at contact (3 cm), after the range has stopped decreasing for 0.5 s, or when a track is lost; episodes shorter than 0.5 s are dropped. Every chase is a per-frame table in centimetres, seconds and radians (pursuer position, heading and speed; target position and velocity; range and bearing), validated for missing frames, inconsistent geometry and implausible speeds. Target velocity for the laws is the causal backward difference over two frames; a centred smoother serves the manoeuvre indicator only. Frames with the target faster than 3 cm/s are called informative.

4.2 The laws and the observation model

With pursuer heading ψ, bearing error b (the angle from the heading to the line of sight) and line-of-sight angle λ, the commanded heading change at frame s is $g\, b_{s-d}$ for pure pursuit, $g\, b^{\tau}_{s-d}$ for lead pursuit, where b^{τ} is the bearing error to the target's position extrapolated by τ seconds at its current velocity, and $N\, \Delta\lambda_{s-d}$ for proportional navigation, with d the delay in frames, g the gain shared by the two pursuit laws and N the navigation constant. The observed increment

is $\Delta\psi_s$ = command + residual. The residual r_s carries a first-order autoregressive term with one shared coefficient ρ, and the innovation $e_s = r_s - \rho r_{s-1}$ is a two-component mixture, a core Gaussian of scale σ and an outlier Gaussian of weight ϵ and scale σ_{out}, which absorbs single-frame tracking glitches. All of d, ρ, σ, ϵ and σ_{out} are shared by every model, including the persistence null (command zero), so that comparisons concern the laws alone.

4.3 Switching and inference

A hidden Markov model over the law index carries either a fixed transition matrix or transition probabilities that depend through a multinomial logistic link on the range, the closing speed and an indicator raised for six frames after a target manoeuvre (a heading change above 60° over three frames while moving faster than 3 cm/s, or a speed change above 15 cm/s). Fitting is by expectation-maximisation: the expectation step runs forward-backward over all chases of a fit in one padded batch, the maximisation step updates the transition parameters in closed form (or by a weighted multinomial logistic regression for the input-dependent model), the gain and N by weighted least squares on the quasi-differenced regressors, the lead time by a bounded search, ρ by a weighted regression of the residual on its lag, and the mixture parameters in closed form; the iterations stop when the log-likelihood improves by less than one part in a million of its value or after 200 iterations (the replication set's switching fit reached the cap, with the fit report recording it), with restarts from perturbed starting points. Single-law models are the same fit with the posterior pinned to one law. The delay is profiled over zero to eight frames at every integer value on the common support that starts after the largest delay, and the best delay is fixed for the dataset. The generic comparator is a switching model with the same number of states whose state k predicts the increment from an intercept and the previous increment, $a_k + b_k \Delta\psi_{s-1}$, with the same shared noise model; a comparator with a free scale per state is rewarded for the frames on which a tracked heading repeats exactly (pixel quantisation, duplicated tracker output) rather than for the steering, and is not used.

Posterior law probabilities at every frame come from the forward-backward pass and decoded paths from the Viterbi algorithm. Parameter intervals come from a chase-level bootstrap stratified by animal, every refit starting from the fitted model with the full settings. Model comparison uses the held-out log-likelihood per frame on the common support, summed per chase, with chase-level bootstrap intervals of the difference from the best single law, on all frames and on informative frames; the per-frame terms of the switching models are the forward predictive log-likelihoods, which sum to the marginal likelihood.

4.4 The recovery study

A simulator with the same observation model as the inference, plus target behaviours the inference does not know (constant velocity with random turns, an evader that turns away below a range threshold, a cricket-like jumper) and scripted switching (at a fixed time, by a Markov chain, triggered by range or by a target manoeuvre), generated a suite of chases exported in the common format. Even replicates fit the models, odd replicates are held out. The study reports law classification by held-out likelihood at four lengths and four noise scales, switching detection against a null set at the 95th percentile of the switching gap on single-law chases with and without tracking noise, false detection on misspecified targets, switch-time error, interval coverage for τ, N, the gain and the delay, posterior calibration, and the limits above which recovery holds. A realistic block matches the gains, delay, residual autocorrelation, tracking noise and glitches of the real recordings; its results are reported as limits in Results.

4.5 Real-data protocol

Every design choice, the thresholds, the library, the emission model and the delay, was made on the development split of the primary dataset and frozen before the held-out split was scored. The

held-out split was scored once before three preprocessing changes were made on the development split, and those numbers were discarded: a running median on the head yaw was replaced by the flip repair above, because the median left a fifth of heading increments exactly zero; the seven untracked clips were excluded; and the comparator was given the noise model of the law models. The held-out numbers reported are from the single pass after these changes. The replication and conspecific sets were fitted once with the frozen structure, each profiling its own delay because frame rate and tracking differ.

The null for switching is a parametric bootstrap at the dataset level: the best single law from the development fit is replayed along the recorded target paths of both splits with the recorded speed profiles and start, with tracking noise of 0.2 cm on positions and 0.05 rad on the heading, refitted on the replayed development set and scored on the replayed held-out set, 200 times; the observed gap is compared with that distribution.

State roles. A state of a switching fit is a steering state when its fitted response to the target is at least 0.05 per frame (the gain for pure and lead pursuit, the navigation constant for proportional navigation) and a non-steering state otherwise; the reported posterior is the steering probability, the posterior mass of the steering states. Within-state regressions of the heading increment on the lagged bearing over the decoded frames confirm the roles: the slope is 0.16, 0.24 and 0.04 in the pure-pursuit state (correlations 0.71, 0.92 and 0.37) and 0.00, 0.01 and 0.00 in the proportional-navigation state (correlations 0.04, 0.09 and -0.05), for the primary, replication and mouse-and-mouse sets.

Follow-up. After the freeze, the library was rerun with an explicit persistence state (zero predicted increment) as a fourth state and the comparator with four states, on the same splits and with the same steps, as a pre-declared follow-up reported after the frozen analysis.

4.6 Optimal play and the two-player test

In the simple-motion pursuit game both players move at constant speed and choose their heading freely; when the pursuer is faster its optimal strategy holds the line of sight at a constant bearing and the evader runs straight away along the line of sight, and the value is the capture time $(r_0 - \ell)/(v_P - v_E)$ for initial range r_0 and capture radius ℓ [1, 2]. The constant-bearing heading given the geometry is $\psi^* = \lambda + \arcsin(u_\perp / v)$, with $u_\perp$ the target's velocity across the line of sight and v the pursuer's speed. The distance from optimal play of a chase is the time average over its frames of $|\psi_{s+d+1} - \psi^*_s|$, the gap between the heading the pursuer adopted after its delay and the heading the optimal strategy prescribed, with a block bootstrap over frames within the chase and chases within animals; the same quantity against the pure-pursuit heading is reported beside it. The optimal capture time is the first time at which a pursuer running at the observed speed profile on the constant-bearing course against the recorded target path brings the range below the capture radius; it is compared with the observed capture time for chases ending in contact.

The two-player test fits a logistic regression of target-manoeuvre onsets, frame by frame, on the range, the closing speed and the pursuer's filtered steering probability at the previous frame (the forward pass alone, which uses no observation after that frame; the smoothed posterior would carry the pursuer's response to the manoeuvre back into its state before it), with chase-level cluster bootstrap intervals. The null circularly shifts the steering covariate within each chase; because the shift keeps each chase's mean, the null distribution carries the between-chase association, and the test statistic is the distance of the observed coefficient from the centre of the null, with a two-sided permutation p-value. For mouse targets the evader library (pure evasion, away from the pursuer's current position; lead evasion, away from its extrapolated position; straight flight, no commanded turn) is fitted to the chased mouse's steering with the same machinery; the pursuer's steering probability is cross-correlated with the evader's lead-evasion

probability against a chase-shuffle null, and the joint occupancy of pursuer state by evader mode is tabulated by fifth of the chase.

4.7 Sensitivity analyses

The replication set is scored leave-one-animal-out: every model refitted on two of the three animals at the frozen delay and scored on the third, with the per-chase scores pooled over folds. The primary dataset is rerun at an arena-fit scale, pixels per centimetre along x and y chosen so that the extent of the tracked head positions between the 0.5th and 99.5th percentiles over all clips matches the stated arena (24.6 by 20.4 px/cm against the deposit's 27), with the head yaw corrected for the anisotropy, at the frozen delay.

4.8 Software

The analysis is one Python package with pinned requirements, fixed seeds, tests, and a command runner that regenerates the synthetic suite, the recovery study, the fits, the evaluation, the tables, the figures and this document; tables and figures regenerate from the committed derived tables without the raw recordings.

Data and code

Code, derived tables, result files and figures are at `https://github.com/eigenreza/markov-switching-pursuit` under the MIT licence. The raw recordings are the public deposits of their authors [27–29]; tables derived from the MABe22 recordings carry that dataset's CC BY-NC-SA licence. Every number in this paper is read from a committed result file by the build script, and a test fails if the paper's tables and the result files disagree.

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A Supplementary tables and figures

Table 7: All columns of Table 3. Fitted parameters of the switching model on the development sets with chase-level bootstrap intervals, and the switching gap against its parametric-bootstrap null.

dataset	delay	delay ms	gain	lo	hi
cricket, primary	1	33.3	0.188	0.152	0.292
cricket, replication	1	35.0	0.238	0.215	0.260
mouse	2	66.7	0.221	0.217	0.227

dataset	lead time	lo	hi
cricket, primary	0.001	0.000	0.344
cricket, replication	0.037	0.000	0.189
mouse	0.225	0.224	0.237

dataset	N	lo	hi
cricket, primary	-0.017	-0.067	0.112
cricket, replication	0.046	-0.014	0.114
mouse	-0.003	-0.004	-0.001

dataset	noise	lo	hi	residual AR	outlier
cricket, primary	0.048	0.045	0.051	0.538	0.032
cricket, replication	0.078	0.075	0.081	0.223	0.040
mouse	0.014	0.014	0.014	0.950	0.017

dataset	stay	gap	null q95	null p
cricket, primary	0.937	0.023	-0.001	<0.005
cricket, replication	0.777	0.026	0.000	<0.005
mouse	0.824	0.077	-0.001	<0.005

Recovery sweeps. The full classification, detection, switch-time, parameter-recovery and calibration tables of the recovery study, the realistic-regime block, and the pooled fits across laws are in the result files `results/recovery/` of the repository, together with the limits note that states the shortest chase and the highest noise at which each target holds.

Chase figures. Per-frame posteriors for the 300 longest held-out chases of each dataset are in `results/fits/<dataset>/posteriors_frames.csv`; the results browser of the repository's demo draws any of them.

Table 8: All columns of Table 4. Posterior summaries on the held-out chases: steering probability, decoded state shares and switches. Roles: S steering, N non-steering, for the states pure pursuit, lead pursuit and proportional navigation, then persistence where present.

dataset	p steer	p steer, moving	steer path	frac pure	frac lead
cricket, primary	0.404	0.388	0.226	0.280	0.002
cricket, replication	0.250	0.243	0.146	0.044	0.068
mouse	0.292	0.272	0.229	0.242	0.006

dataset	frac pn	switches	with switch	roles
cricket, primary	0.718	0.131	0.105	S S N
cricket, replication	0.888	1.316	0.508	S S N
mouse	0.751	0.832	0.500	S S N

Table 9: All columns of Table 5. Distance from optimal play and the two-player test.

dataset	dist opt (deg)	q25	q75	dist, moving	dist pure
cricket, primary	22.7	13.6	41.2	31.8	20.2
cricket, replication	30.7	20.0	44.9	36.0	25.3
mouse	63.6	45.0	87.9	75.2	44.0

dataset	contact	capture ratio	man. coef	lo	hi
cricket, primary	5	1.206	-0.217	-1.875	1.198
cricket, replication	55	1.154	0.084	-0.579	0.598
mouse	451	1.125	0.334	0.229	0.438

dataset	null mean	within	perm. p	onsets	xcorr 0
cricket, primary	0.335	-0.552	0.275	33	n/a
cricket, replication	0.229	-0.146	0.555	268	n/a
mouse	0.318	0.016	0.693	5,491	0.011

dataset	xcorr peak	peak lag
cricket, primary	n/a	n/a
cricket, replication	n/a	n/a
mouse	0.011	1

Table 10: All columns of Table 6. The post-freeze follow-up with a persistence state, against the frozen three-law analysis. Roles: S steering, N non-steering, for the states pure pursuit, lead pursuit and proportional navigation, then persistence where present.

dataset	delay	gain	lead time
cricket, primary, 3 laws	1	0.188	0.001
cricket, primary, + persistence	1	0.280	0.000
cricket, replication, 3 laws	1	0.238	0.037
cricket, replication, + persistence	1	0.248	0.001
mouse, 3 laws	2	0.221	0.225
mouse, + persistence	2	0.227	0.228

dataset	N	lo	hi
cricket, primary, 3 laws	-0.017	-0.067	0.112
cricket, primary, + persistence	-0.025	-0.073	0.117
cricket, replication, 3 laws	0.046	-0.014	0.114
cricket, replication, + persistence	0.158	0.029	0.262
mouse, 3 laws	-0.003	-0.004	-0.001
mouse, + persistence	-0.547	-1.143	-0.539

dataset	gap	lo	hi	comparator	null p
cricket, primary, 3 laws	0.023	0.017	0.029	0.028	<0.005
cricket, primary, + persistence	0.028	0.020	0.035	0.030	0.032
cricket, replication, 3 laws	0.026	0.018	0.033	0.006	<0.005
cricket, replication, + persistence	0.029	0.022	0.037	0.012	0.032
mouse, 3 laws	0.077	0.075	0.079	0.296	<0.005
mouse, + persistence	0.083	0.081	0.085	0.271	0.048

dataset	p steer	frac pure	frac pn	persist	roles
cricket, primary, 3 laws	0.404	0.280	0.718	0.000	S S N
cricket, primary, + persistence	0.257	0.005	0.569	0.325	S S N N
cricket, replication, 3 laws	0.250	0.044	0.888	0.000	S S N
cricket, replication, + persistence	0.543	0.062	0.384	0.526	S S S N
mouse, 3 laws	0.292	0.242	0.751	0.000	S S N
mouse, + persistence	0.367	0.276	0.027	0.689	S S S N

Table 11: Replication set scored leave-one-animal-out: held-out log-likelihood per frame minus the best single law, pooled over three folds, with chase-level bootstrap intervals.

frames	model	per frame	diff	lo	hi	chases	frames
all	null	0.9024	-0.0121	-0.0166	-0.0071	177	7,537
all	single pure pursuit	0.9145	0.0000	0.0000	0.0000	177	7,537
all	single lead pursuit	0.9145	0.0000	-0.0010	0.0010	177	7,537
all	single proportional navigation	0.9047	-0.0098	-0.0150	-0.0048	177	7,537
all	switching plain	0.9362	0.0218	0.0141	0.0296	177	7,537
all	switching input	0.9427	0.0282	0.0195	0.0367	177	7,537
all	switching ar	0.9195	0.0050	-0.0019	0.0125	177	7,537
informative	null	0.8796	-0.0101	-0.0166	-0.0036	151	2,831
informative	single pure pursuit	0.8896	0.0000	0.0000	0.0000	151	2,831
informative	single lead pursuit	0.8892	-0.0005	-0.0020	0.0010	151	2,831
informative	single proportional navigation	0.8809	-0.0088	-0.0169	-0.0009	151	2,831
informative	switching plain	0.9124	0.0228	0.0106	0.0352	151	2,831
informative	switching input	0.9180	0.0283	0.0137	0.0437	151	2,831
informative	switching ar	0.8980	0.0084	-0.0020	0.0186	151	2,831

Table 12: Primary dataset at the deposit scale and at the arena-fit scale.

scale	dev chases	held chases	held moving	delay
deposit scale	210	275	1,022	1
arena-fit scale	227	311	1,312	1

scale	gain	lo	hi	N
deposit scale	0.188	0.152	0.292	-0.017
arena-fit scale	0.168	0.142	0.326	-0.025

scale	gap	lo	hi	comparator	null p
deposit scale	0.023	0.017	0.029	0.028	<0.005
arena-fit scale	0.021	0.016	0.026	0.033	0.032

scale	p steer
deposit scale	0.404
arena-fit scale	0.400

Table 13: What predicts switching. Held-out log-likelihood per frame of input-dependent minus fixed transitions, with chase-level bootstrap intervals, and its share of the switching gap; steering probability from 0.5 to 0.1 s before and from 0.1 to 0.5 s after a target manoeuvre, with the prediction from chase phase alone.

dataset	input gain	lo	hi	gap	share	chases	before	after	phase before	phase after	change	phase change
mouse and cricket (primary)	0.0078	0.0034	0.0119	0.0231	0.3390	17	0.4756	0.4691	0.3446	0.3609	-0.0064	0.0163
mouse and cricket (replication)	0.0096	0.0067	0.0128	0.0255	0.3773	114	0.2884	0.2731	0.2265	0.2292	-0.0153	0.0027
mouse and mouse	0.0021	0.0019	0.0023	0.0768	0.0270	3,635	0.3622	0.2391	0.3166	0.2380	-0.1231	-0.0786

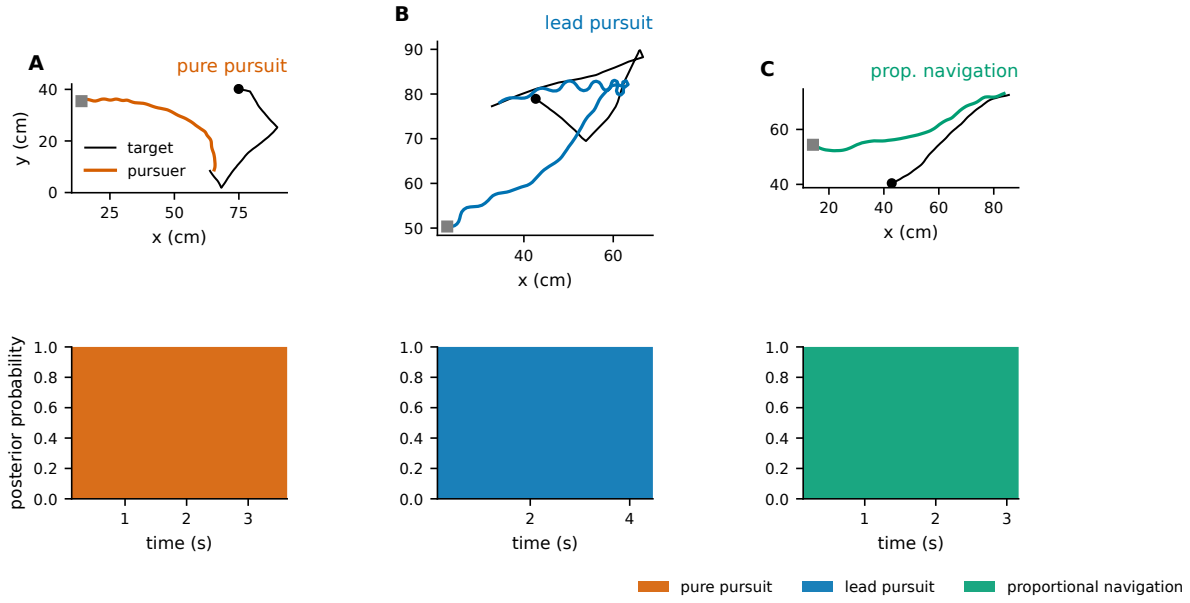


Figure 10: Simulated chases, one per law, with the inferred law posterior underneath. The pursuer (grey) chases a turning target (black); the colour of the pursuer's path is the decoded law.